

Probability

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Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
```

```
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif"))
)

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

Most of the content in this chapter should be review from UC Davis Epi 202.

1 Core properties of probabilities

1.1 Defining probabilities

Definition 1.1 (Probability measure). A **probability measure**, often denoted $\Pr()$ or $P()$, is a function whose domain is a σ -algebra^a of possible outcomes, $\mathcal{S}$, and which satisfies the following properties:

1. For any statistical event $A \in \mathcal{S}$, $\Pr(A) \geq 0$.
2. The probability of the union of all outcomes ($\Omega \stackrel{\text{def}}{=} \cup \mathcal{S}$) is 1:

$$\Pr(\Omega) = 1$$

3. The probability of the union of countably many mutually disjoint events $A_1, A_2, \dots$ (where $A_i \cap A_j = \emptyset$ for all $i \neq j$) is equal to the sum of their probabilities (*countable additivity* or *sigma-additivity*):

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

^a<https://en.wikipedia.org/wiki/%CE%A3-algebra>

Property 3 (*countable additivity*) is stronger than *finite additivity*, which only requires

$$\Pr(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n \Pr(A_i)$$

for every finite collection of mutually disjoint events. Countable additivity implies finite additivity (set $A_{n+1} = A_{n+2} = \dots = \emptyset$ in property 3, using $\Pr(\emptyset) = 0$), but not vice versa: there exist set functions that satisfy finite additivity but fail countable additivity (see Wikipedia: Sigma-additive set function — An additive function which is not σ -additive¹). Requiring countable additivity enables results such as the continuity of probability (if $A_1 \supseteq A_2 \supseteq \dots$ with $\bigcap_i A_i = \emptyset$, then $\Pr(A_i) \rightarrow 0$) and underpins the Theorem 1.4 for countable partitions.

Theorem 1.1 (Probability of a subset's intersection). *If A and B are statistical events and $A \subseteq B$, then $\Pr(A \cap B) = \Pr(A)$.*

i Proof

Proof. Left to the reader for now. □

Theorem 1.2 (An event and its complement sum to 1).

$$\Pr(A) + \Pr(\neg A) = 1$$

i Proof

Proof. By properties 2 and 3 of Definition 1.1. □

Corollary 1.1 (Complement rule).

$$\Pr(\neg A) = 1 - \Pr(A)$$

i Proof

Proof. By Theorem 1.2 and algebra. □

Corollary 1.2 (Complement rule in probability (π) notation). *If the probability of an outcome A is $\Pr(A) = \pi$, then the probability that A does not occur is:*

¹https://en.wikipedia.org/wiki/Sigma-additive_set_function#An_additive_function_which_is_not_%CF%83-additive

$$\Pr(\neg A) = 1 - \pi$$

i Proof

Proof. Using Corollary 1.1:

$$\begin{aligned}\Pr(\neg A) &= 1 - \Pr(A) \\ &= 1 - \pi\end{aligned}$$

□

1.2 Conditional probability

Definition 1.2 (Conditional probability). For two events A and B with $\Pr(B) > 0$, the **conditional probability** of A given B , denoted $\Pr(A \mid B)$, is:

$$\Pr(A \mid B) \stackrel{\text{def}}{=} \frac{\Pr(A \cap B)}{\Pr(B)}$$

Theorem 1.3 (Law of conditional probability). For any two events A and B with $\Pr(B) > 0$:

$$\Pr(A \cap B) = \Pr(A \mid B) \cdot \Pr(B)$$

i Proof

Proof. Rearranging Definition 1.2:

$$\begin{aligned}\Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} \\ \Pr(A \cap B) &= \Pr(A \mid B) \cdot \Pr(B)\end{aligned}$$

□

Exm

Example 1.1 (Applying the law of conditional probability). Suppose 30% of adults exercise regularly ($\Pr(E) = 0.30$), and among adults who exercise regularly, 60% have low blood pressure ($\Pr(L \mid E) = 0.60$).

Then the probability that a randomly selected adult both exercises regularly and has low blood pressure is:

$$\begin{aligned}\Pr(L \cap E) &= \Pr(L \mid E) \cdot \Pr(E) \\ &= 0.60 \cdot 0.30 \\ &= 0.18\end{aligned}$$

Theorem 1.4 (Law of total probability). *If $B_1, B_2, \dots$ is a countable partition of the sample space (i.e., countably many mutually exclusive events whose union is the entire sample space), then for any event A :*

$$\Pr(A) = \sum_{i=1}^{\infty} \Pr(A \mid B_i) \cdot \Pr(B_i)$$

i Proof

Proof. Since $B_1, B_2, \dots$ partition the sample space, the events $A \cap B_1, A \cap B_2, \dots$ are mutually exclusive and their union is A . By property 3 of Definition 1.1 (countable additivity), and then by Theorem 1.3:

$$\begin{aligned} \Pr(A) &= \sum_{i=1}^{\infty} \Pr(A \cap B_i) \\ &= \sum_{i=1}^{\infty} \Pr(A \mid B_i) \cdot \Pr(B_i) \end{aligned}$$

□

Theorem 1.5 (Bayes' theorem). *For any two events A and B with $\Pr(A) > 0$ and $\Pr(B) > 0$:*

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \cdot \Pr(A)}{\Pr(B)}$$

i Proof

Proof. Apply Definition 1.2 to both $\Pr(A \mid B)$ and $\Pr(B \mid A)$:

$$\begin{aligned} \Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} \\ &= \frac{\Pr(B \mid A) \cdot \Pr(A)}{\Pr(B)} \end{aligned}$$

The second equality follows from Theorem 1.3 applied to $\Pr(B \cap A) = \Pr(B \mid A) \cdot \Pr(A)$. □

Exm

Example 1.2 (Positive predictive value of a medical test). Suppose a disease test has 99% sensitivity and 99% specificity, and the prevalence of the disease in the population is 7%. Let D be the event “person has the disease” and $+$ be the event “test is positive”. Then:

- $\Pr(+ \mid D) = 0.99$ (sensitivity)
- $\Pr(\neg+ \mid \neg D) = 0.99$ (specificity), so the false positive rate is $\Pr(+ \mid \neg D) = 1 - 0.99 = 0.01$
- $\Pr(D) = 0.07$ (prevalence)

By Bayes' theorem (Theorem 1.5) and the law of total probability (Theorem 1.4):

$$\begin{aligned}
\Pr(D \mid +) &= \frac{\Pr(+ \mid D) \cdot \Pr(D)}{\Pr(+)} \\
&= \frac{\Pr(+ \mid D) \cdot \Pr(D)}{\Pr(+ \mid D) \cdot \Pr(D) + \Pr(+ \mid \neg D) \cdot \Pr(\neg D)} \\
&= \frac{0.99 \cdot 0.07}{0.99 \cdot 0.07 + 0.01 \cdot 0.93} \\
&= \frac{0.0693}{0.0693 + 0.0093} \\
&= \frac{0.0693}{0.0786} \\
&\approx 0.88
\end{aligned}$$

Even with a highly accurate test (99% sensitive and 99% specific), only about 88% of people who test positive actually have the disease, because the disease prevalence is relatively low (7%).

2 Key probability distributions

Some distributions are typically used for outcome models (Table 1); other distributions are typically used for test statistics (Table 2).

Table 1: Distributions typically used for outcome models

Distribution	Uses
Bernoulli	Binary outcomes
Binomial	Sums of Bernoulli outcomes
Poisson	unbounded count outcomes
Geometric	Counts of non-events before an event occurs
Negative binomial	Mixtures of Poisson distributions, counts of non-events until a given number of events occurs
Normal (Gaussian)	Continuous outcomes without a more specific distribution
exponential	Time to event outcomes
Gamma	Time to event outcomes
Weibull	Time to event outcomes
Log-normal	Time to event outcomes

Table 2: Distributions typically used for test statistics

Distribution	Uses
χ^2	Regression comparisons (asymptotic), contingency table independence tests, goodness-of-fit tests
F	Gaussian model comparisons (exact)
Z (standard normal)	Proportions, means, regression coefficients (asymptotic)
T	Means, regression coefficients in Gaussian outcome models (exact)

2.1 The Bernoulli distribution

Definition 2.1 (Bernoulli distribution). The **Bernoulli distribution** family for a random variable X is defined as:

$$\begin{aligned}\Pr(X = x) &= 1_{x \in \{0,1\}} \pi^x (1 - \pi)^{1-x} \\ &= \begin{cases} \pi, & x = 1 \\ 1 - \pi, & x = 0 \end{cases}\end{aligned}$$

2.2 The Poisson distribution



(a) Siméon Denis Poisson



(b) Les Poissons^a

^a<https://youtu.be/UoJxBEQRLd0?t=12>

Figure 1: “Les Poissons”

Exercise 2.1. Define the Poisson distribution.

Solution

Solution 2.1.

Def

Definition 2.2 (Poisson distribution).

$$P(Y = y) = \frac{\mu^y e^{-\mu}}{y!}, y \in \mathbb{N} \quad (1)$$

(see Figure 2)

Exercise 2.2. What is the range of possible values for a Poisson distribution?

Solution

Solution 2.2.

$$\mathcal{R}(Y) = \{0, 1, 2, \dots\} = \mathbb{N}$$

Theorem 2.1 (CDF of Poisson distribution).

$$P(Y \leq y) = e^{-\mu} \sum_{j=0}^{\lfloor y \rfloor} \frac{\mu^j}{j!} \quad (2)$$

(see Figure 3)

```
library(dplyr)
pois_dists <- tibble(
  mu = c(0.5, 1, 2, 5, 10, 20)
) |>
  reframe(
    .by = mu,
    x = 0:30
  ) |>
  mutate(
    `P(X = x)` = dpois(x, lambda = mu),
    `P(X <= x)` = ppois(x, lambda = mu),
    mu = factor(mu)
  )

library(ggplot2)
library(latex2exp)

plot0 <- pois_dists |>
  ggplot(
    aes(
      x = x,
      y = `P(X = x)`,
      fill = mu,
      col = mu
    )
  ) +
  theme(legend.position = "bottom") +
  labs(
    fill = latex2exp::TeX("$\\mu$"),
    col = latex2exp::TeX("$\\mu$"),
    y = latex2exp::TeX("$\\Pr_{\\mu}(X = x)$")
  )

plot1 <- plot0 +
  geom_segment(yend = 0) +
  facet_wrap(~mu)

print(plot1)
```

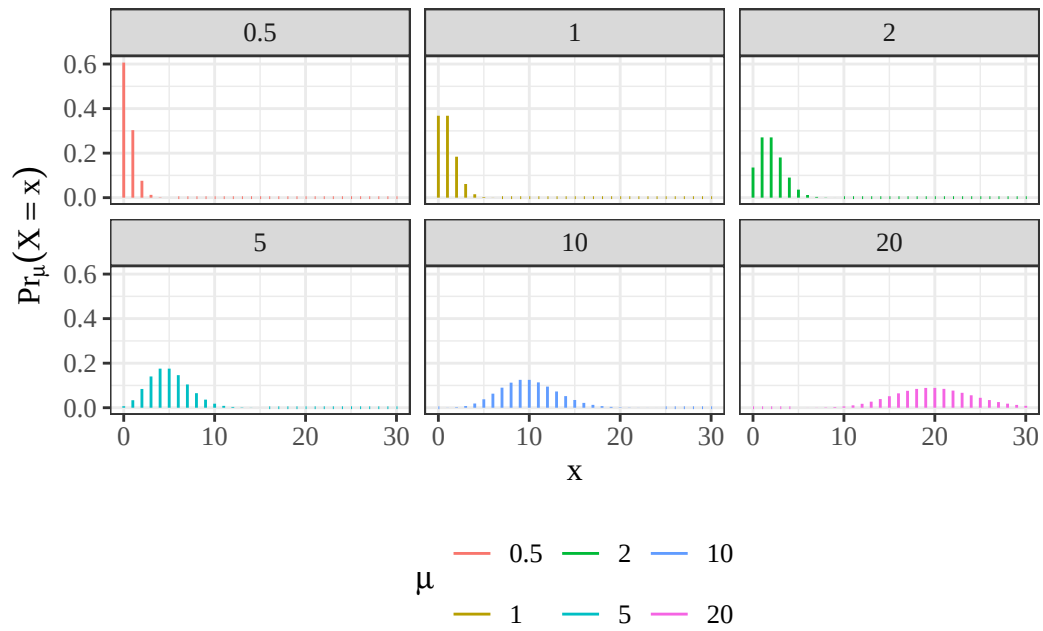


Figure 2: Poisson PMFs, by mean parameter μ

```
library(ggplot2)

plot2 <-
  plot0 +
  geom_step(alpha = 0.75) +
  aes(y = `P(X <= x)`) +
  labs(y = latex2exp::TeX("$\\Pr_{\\mu}(X \\leq x)$"))

print(plot2)
```

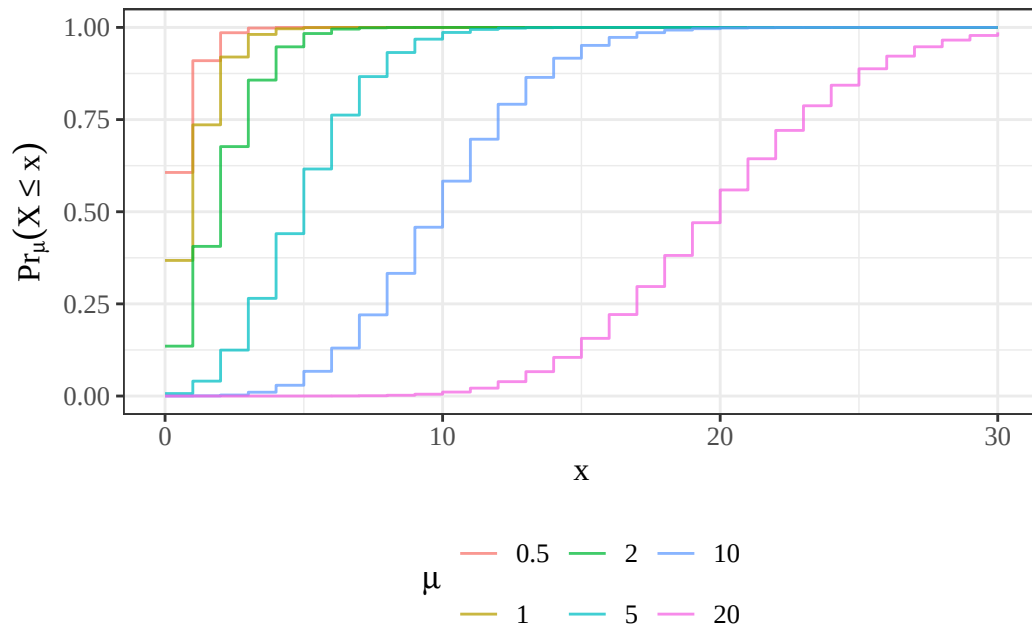


Figure 3: Poisson CDFs

Exercise 2.3 (Poisson distribution functions). Let $X \sim \text{Pois}(\mu = 3.75)$.

Compute:

- $P(X = 4 | \mu = 3.75)$
- $P(X \leq 7 | \mu = 3.75)$
- $P(X > 5 | \mu = 3.75)$

Solution

Solution.

- $P(X = 4) = 0.19378$
- $P(X \leq 7) = 0.962379$
- $P(X > 5) = 0.177117$

Theorem 2.2 (Properties of the Poisson distribution). If $X \sim \text{Pois}(\mu)$, then:

- $E[X] = \mu$
- $\text{Var}(X) = \mu$
- $P(X = x) = \frac{\mu}{x} P(X = x - 1)$
- For $x < \mu$, $P(X = x) > P(X = x - 1)$
- For $x = \mu$, $P(X = x) = P(X = x - 1)$
- For $x > \mu$, $P(X = x) < P(X = x - 1)$
- $\arg \max_x P(X = x) = \lfloor \mu \rfloor$

Exercise 2.4. Prove Theorem 2.2.

Solution

Solution.

$$\begin{aligned} E[X] &= \sum_{x=0}^{\infty} x \cdot P(X = x) \\ &= 0 \cdot P(X = 0) + \sum_{x=1}^{\infty} x \cdot P(X = x) \\ &= 0 + \sum_{x=1}^{\infty} x \cdot P(X = x) \\ &= \sum_{x=1}^{\infty} x \cdot P(X = x) \\ &= \sum_{x=1}^{\infty} x \cdot \frac{\lambda^x e^{-\lambda}}{x!} \\ &= \sum_{x=1}^{\infty} x \cdot \frac{\lambda^x e^{-\lambda}}{x \cdot (x-1)!} && \text{[definition of factorial ("!") function]} \\ &= \sum_{x=1}^{\infty} \frac{\lambda^x e^{-\lambda}}{(x-1)!} \\ &= \sum_{x=1}^{\infty} \frac{(\lambda \cdot \lambda^{x-1}) e^{-\lambda}}{(x-1)!} \\ &= \lambda \cdot \sum_{x=1}^{\infty} \frac{(\lambda^{x-1}) e^{-\lambda}}{(x-1)!} \\ &= \lambda \cdot \sum_{y=0}^{\infty} \frac{(\lambda^y) e^{-\lambda}}{(y)!} && \text{[substituting } y \stackrel{\text{def}}{=} x - 1] \\ &= \lambda \cdot 1 && \text{[because PDFs sum to 1]} \\ &= \lambda \end{aligned}$$

See also <https://statproofbook.github.io/P/poiss-mean>.

For the variance, see <https://statproofbook.github.io/P/poiss-var>.

Accounting for exposure

Definition 2.3 (Exposure magnitude). For many count outcomes, there is some sense of an **exposure magnitude**, such as **population size**, or **duration of observation**, which multiplicatively rescales the expected (mean) count.

Exercise 2.5. What are some examples of exposure magnitudes?

Solution

Solution.

Table 3: Examples of exposure units

outcome	exposure units
disease incidence	number of individuals exposed; time at risk
car accidents	miles driven
worksite accidents	person-hours worked
population size	size of habitat

Exposure units are similar to the number of trials in a binomial distribution, but **in non-binomial count outcomes, there can be more than one event per unit of exposure.**

We can use t to represent continuous-valued exposures/observation durations, and n to represent discrete-valued exposures.

Definition 2.4 (Event rate).

For a count outcome Y with exposure magnitude t , the **event rate** (denoted λ) is defined as the mean of Y divided by the exposure magnitude. That is:

$$\begin{aligned}\mu &\stackrel{\text{def}}{=} \text{E}[Y|T = t] \\ \lambda &\stackrel{\text{def}}{=} \frac{\mu}{t}\end{aligned}\tag{3}$$

Event rate is somewhat analogous to odds in binary outcome models; it typically serves as an intermediate transformation between the mean of the outcome and the linear component of the model. However, in contrast with the odds function, the transformation $\lambda = \mu/t$ is *not* considered part of the Poisson model's link function, and it treats the exposure magnitude covariate differently from the other covariates.

Theorem 2.3 (Transformation function from event rate to mean). *For a count variable with mean μ , event rate λ , and exposure magnitude t :*

$$\mu = \lambda \cdot t\tag{4}$$

Solution

Solution. Start from definition of event rate and use algebra to solve for μ .

Equation 4 is analogous to the inverse-odds function for binary variables.

Theorem 2.4 (No exposure means no expected events). *When the exposure magnitude is 0, there is no opportunity for events to occur:*

$$\text{E}[Y|T = 0] = 0$$

i Proof

Proof.

$$E[Y|T = 0] = \lambda \cdot 0 = 0$$

□

! Important

The exposure magnitude, T , is *similar* to a covariate in linear or logistic regression. However, there is an important difference: in count regression, **there is no intercept corresponding to $E[Y|T = 0]$** . In other words, this model assumes that if there is no exposure, there can't be any events.

Theorem 2.5 (Exposure is additive on the log scale). *If $\mu = \lambda \cdot t$, then:*

$$\log \mu = \log \lambda + \log t$$

Definition 2.5 (Offset). When the linear component of a model involves a term without an unknown coefficient, that term is called an **offset**.

Theorem 2.6 (Sum of independent Poisson random variables). *If X and Y are independent Poisson random variables with means μ_X and μ_Y , their sum, $Z = X + Y$, is also a Poisson random variable, with mean $\mu_Z = \mu_X + \mu_Y$.*

i Proof

Proof. See https://web.stanford.edu/class/archive/cs/cs109/cs109.1206/lectureNotes/LN12_independent_rvs.pdf, Example 3. □

2.3 The Negative-Binomial distribution

Definition 2.6 (Negative binomial distribution).

$$P(Y = y) = \frac{\mu^y}{y!} \cdot \frac{\Gamma(\rho + y)}{\Gamma(\rho) \cdot (\rho + \mu)^y} \cdot \left(1 + \frac{\mu}{\rho}\right)^{-\rho}$$

where ρ is an overdispersion parameter and $\Gamma(x) = (x - 1)!$ for integers x .

You don't need to memorize or understand this expression.

As $\rho \rightarrow \infty$, the second term converges to 1 and the third term converges to $\exp\{-\mu\}$, which brings us back to the Poisson distribution.

Theorem 2.7 (Mean and variance of the negative binomial distribution). *If $Y \sim \text{NegBin}(\mu, \rho)$, then:*

- $E[Y] = \mu$
- $\text{Var}(Y) = \mu + \frac{\mu^2}{\rho} > \mu$

2.4 Weibull Distribution

$$p(t) = \alpha \lambda x^{\alpha-1} e^{-\lambda x^\alpha}$$

$$\lambda(t) = \alpha \lambda x^{\alpha-1}$$

$$S(t) = e^{-\lambda x^\alpha}$$

$$E(T) = \Gamma(1 + 1/\alpha) \cdot \lambda^{-1/\alpha}$$

When $\alpha = 1$ this is the exponential. When $\alpha > 1$ the hazard is increasing and when $\alpha < 1$ the hazard is decreasing. This provides more flexibility than the exponential.

We will see more of this distribution later.

3 Characteristics of probability distributions

3.1 Probability density function

Definition 3.1 (probability density). If X is a continuous random variable, then the **probability density** of X at value x , denoted $f(x)$, $f_X(x)$, $p(x)$, $p_X(x)$, or $p(X = x)$, is defined as the limit of the **probability** (mass) that X is in an interval around x , divided by the width of that interval, as that width reduces to 0.

$$f(x) \stackrel{\text{def}}{=} \lim_{\Delta \rightarrow 0} \frac{P(X \in [x, x + \Delta])}{\Delta}$$

See also Rothman et al. (2021) (Chapter 22, p. 535) and https://en.wikipedia.org/wiki/Probability_density_function#Formal_definition

Definition 3.2 (Cumulative distribution function (CDF)). For a random variable X , its population CDF is

$$F(t) = \Pr(X \leq t), \quad t \in \mathbb{R}.$$

Definition 3.3 (Quantile function (population inverse CDF)). For a random variable X with **cumulative distribution function (CDF)** F , its population quantile function (generalized inverse of F) is

$$Q(p) = \inf\{t : F(t) \geq p\}, \quad 0 < p \leq 1.$$

Theorem 3.1 (Density function is derivative of CDF). *The density function $f(t)$ or $p(T = t)$ for a random variable T at value t is equal to the derivative of the cumulative probability function $F(t) \stackrel{\text{def}}{=} P(T \leq t)$; that is:*

$$f(t) \stackrel{\text{def}}{=} \frac{\partial}{\partial t} F(t)$$

Theorem 3.2 (Density functions integrate to 1). *For any density function $f(x)$,*

$$\int_{x \in \mathcal{R}(X)} f(x) dx = 1$$

3.2 Hazard function

Definition 3.4 (Hazard function, hazard rate, hazard rate function).

The **hazard function**, **hazard rate**, **hazard rate function**, for a random variable T at value t , typically denoted as $h(t)$ ² or $\lambda(t)$, ³ is the conditional density^a of T at t , given $T \geq t$. That is:

$$\lambda(t) \stackrel{\text{def}}{=} p(T = t | T \geq t)$$

If T represents the time at which an event occurs, then $\lambda(t)$ is the probability that the event occurs at time t , given that it has not occurred prior to time t .

The name “hazard” carries a connotation that the event is undesirable — death, relapse, equipment failure, etc. When the event in question is neutral or desirable (recovery, conception, graduation, response to treatment), the same quantity $\lambda(t)$ is often called the **event incidence rate** instead. This is parallel to the convention that conditional probabilities of undesirable events are called **risks**, while the same conditional probabilities for neutral/desirable events are simply called **probabilities**. The math is identical; only the name changes with the valence of the event.

^a[probability.qmd#def-pdf](#)

Table 4: Probability distribution functions

Name	Symbols	Definition
Probability density function (PDF)	$f(t), p(t)$	$p(T = t)$
Cumulative distribution function (CDF)	$F(t), P(t)$	$P(T \leq t)$
Survival function	$S(t), \bar{F}(t)$	$P(T > t)$
Hazard function	$\lambda(t), h(t)$	$p(T = t T \geq t)$
Cumulative hazard function	$\Lambda(t), H(t)$	$\int_{u=-\infty}^t \lambda(u) du$
Log-hazard function	$\eta(t)$	$\log\{\lambda(t)\}$

$$f(t) \xleftarrow{\frac{-S'(t)}{S(t)\lambda(t)}} S(t) \xleftarrow{\exp\{-\Lambda(t)\}} \Lambda(t) \xleftarrow{\int_{u=0}^t \lambda(u) du} \lambda(t) \xleftarrow{\exp\{\eta(t)\}} \eta(t)$$

$$f(t) \xrightarrow{\frac{f(t)/\lambda(t)}{\int_{u=t}^{\infty} f(u) du}} S(t) \xrightarrow{-\log S(t)} \Lambda(t) \xrightarrow{\Lambda'(t)} \lambda(t) \xrightarrow{\log\{\lambda(t)\}} \eta(t)$$

³for example in Dobson and Barnett (2018), Vittinghoff et al. (2012), Klein and Moeschberger (2003), and Kleinbaum and Klein (2012)

³for example, in Rothman et al. (2021) and Kalbfleisch and Prentice (2011)

3.3 Expectation

Definition 3.5 (Expectation, expected value, population mean). The **expectation**, **expected value**, or **population mean** of a *continuous* random variable X , denoted $E[X]$, $\mu(X)$, or μ_X , is the weighted mean of X 's possible values, weighted by the **probability density function** of those values:

$$E[X] = \int_{x \in \mathcal{R}(X)} x \cdot p(X = x) dx$$

The **expectation**, **expected value**, or **population mean** of a *discrete* random variable X , denoted $E[X]$, $\mu(X)$, or μ_X , is the mean of X 's possible values, weighted by the probability mass function of those values:

$$E[X] = \sum_{x \in \mathcal{R}(X)} x \cdot P(X = x)$$

(c.f. https://en.wikipedia.org/wiki/Expected_value)

Theorem 3.3 (Expectation of the Bernoulli distribution). *The expectation of a Bernoulli random variable with parameter π is:*

$$E[X] = \pi$$

i Proof

Proof.

$$\begin{aligned} E[X] &= \sum_{x \in \mathcal{R}(X)} x \cdot P(X = x) \\ &= \sum_{x \in \{0,1\}} x \cdot P(X = x) \\ &= (0 \cdot P(X = 0)) + (1 \cdot P(X = 1)) \\ &= (0 \cdot (1 - \pi)) + (1 \cdot \pi) \\ &= 0 + \pi \\ &= \pi \end{aligned}$$

□

Theorem 3.4 (Expectation of time-to-event variables). *If T is a non-negative random variable, then:*

$$\mu(T|\tilde{X} = \tilde{x}) = \int_{t=0}^{\infty} S(t) dt$$

i Proof

Proof. We prove the continuous case, in which T has a density f . The result follows from applying Tonelli's theorem (hypothesis (a) of Fubini–Tonelli^a) to the function $g(t, u) = f(u) \cdot \mathbb{1}(0 \leq t \leq u)$ on the product space $[0, \infty) \times [0, \infty)$: g is nonnegative everywhere and vanishes outside the (unbounded) triangular region $D = \{(t, u) : 0 \leq t \leq u < \infty\}$, so the

iterated integrals over D are exchangeable.

Since $f(u) \geq 0$, hypothesis (a) of Fubini–Tonelli^b (the nonnegative case, **Tonelli’s theorem**) applies, and we may exchange the order of integration over D :

$$\begin{aligned} \mathbb{E}[T] &= \int_{u=0}^{\infty} u f(u) du \\ &= \int_{u=0}^{\infty} \left(\int_{t=0}^u 1 dt \right) f(u) du \\ &= \int_{u=0}^{\infty} \int_{t=0}^u f(u) dt du \\ &= \int_{t=0}^{\infty} \int_{u=t}^{\infty} f(u) du dt \\ &= \int_{t=0}^{\infty} \mathbb{P}(T > t) dt \\ &= \int_{t=0}^{\infty} S(t) dt. \end{aligned}$$

□

^a[math-prereqs.qmd#thm-fubini-tonelli](#)

^b[math-prereqs.qmd#thm-fubini-tonelli](#)

Exm

Example 3.1 (Mean of an exponential random variable via survival function). Let $T \sim \text{Exponential}(\lambda)$, so $S(t) = e^{-\lambda t}$ for $t \geq 0$. By Theorem 3.4:

$$\begin{aligned} \mathbb{E}[T] &= \int_0^{\infty} S(t) dt \\ &= \int_0^{\infty} e^{-\lambda t} dt \\ &= \left[-\frac{1}{\lambda} e^{-\lambda t} \right]_0^{\infty} \\ &= \frac{1}{\lambda}, \end{aligned}$$

confirming the standard result $\mathbb{E}[T] = 1/\lambda$.

Theorem 3.5 (Law of the Unconscious Statistician (LOTUS)). *For any function g of a discrete random variable X :*

$$\mathbb{E}[g(X)] = \sum_{x \in \mathcal{R}(X)} g(x) \cdot \mathbb{P}(X = x)$$

i Proof

Proof. Let $Y = g(X)$. By Definition 3.5 applied to Y :

$$\begin{aligned}
E[g(X)] &= E[Y] \\
&= \sum_{y \in \mathcal{R}(Y)} y \cdot P(Y = y) \\
&= \sum_{y \in \mathcal{R}(Y)} y \cdot P(g(X) = y) \\
&= \sum_{y \in \mathcal{R}(Y)} y \cdot \sum_{\substack{x \in \mathcal{R}(X) \\ g(x)=y}} P(X = x) \\
&= \sum_{x \in \mathcal{R}(X)} g(x) \cdot P(X = x)
\end{aligned}$$

where the last equality follows by rearranging the double sum, grouping each term x by its image $y = g(x)$. $\square$

LOTUS says that to compute $E[g(X)]$, we do not need to first find the distribution of $g(X)$; we can compute the expectation directly using the distribution of X .

For a *continuous* random variable X with density $p(X = x)$, the analogous formula is:

$$E[g(X)] = \int_{x \in \mathcal{R}(X)} g(x) \cdot p(X = x) dx$$

Exm

Example 3.2 (Expected value of X^2 for a Bernoulli variable). Let $X \sim \text{Ber}(\pi)$. By LOTUS (Theorem 3.5):

$$\begin{aligned}
E[X^2] &= \sum_{x \in \{0,1\}} x^2 \cdot P(X = x) \\
&= 0^2 \cdot P(X = 0) + 1^2 \cdot P(X = 1) \\
&= 0^2 \cdot (1 - \pi) + 1^2 \cdot \pi \\
&= 0 + \pi \\
&= \pi
\end{aligned}$$

Definition 3.6 (Conditional expectation). **Discrete case.** Let X and Y be jointly distributed discrete random variables. The **conditional probability mass function** of Y given $X = x$ (for values of x with $P(X = x) > 0$) is:

$$P(Y = y \mid X = x) \stackrel{\text{def}}{=} \frac{P(X = x, Y = y)}{P(X = x)}$$

The **conditional expectation** of Y given $X = x$ is:

$$E[Y \mid X = x] \stackrel{\text{def}}{=} \sum_{y \in \mathcal{R}(Y)} y \cdot P(Y = y \mid X = x)$$

Continuous case. Let X and Y be jointly distributed continuous random variables with joint density $p(X = x, Y = y)$ and marginal density $p(X = x)$. The **conditional probability density function** of Y given $X = x$ (for values of x with $p(X = x) > 0$) is:

$$p(Y = y \mid X = x) \stackrel{\text{def}}{=} \frac{p(X = x, Y = y)}{p(X = x)}$$

The **conditional expectation** of Y given $X = x$ is:

$$\mathbb{E}[Y \mid X = x] \stackrel{\text{def}}{=} \int_{y \in \mathcal{R}(Y)} y \cdot p(Y = y \mid X = x) dy$$

Conditional expectation function. The **conditional expectation function** $\mathbb{E}[Y \mid X]$ is the function (and hence random variable) of X obtained by evaluating $\mathbb{E}[Y \mid X = x]$ at X ; that is, $\mathbb{E}[Y \mid X] = g(X)$ where $g(x) \stackrel{\text{def}}{=} \mathbb{E}[Y \mid X = x]$.

3.4 Fubini–Tonelli for expectations

The Riemann version of Fubini’s theorem⁴, stated in the math-prereqs chapter, lets us switch the order of integration for continuous integrands on simple regions. For expectations against probability measures we use its measure-theoretic form⁵, which holds on any σ -finite measure space. The σ -finiteness hypothesis is automatic for probability measures (every probability measure is finite, hence σ -finite), so Fubini–Tonelli⁶ yields the corollary below directly.

Corollary 3.1 (Joint-distribution form (without independence; corollary of Fubini–Tonelli)). *Let (X, Y) be jointly distributed random variables whose joint distribution has a density $f_{X,Y}$ with respect to a product of σ -finite reference measures $\mu_X \otimes \mu_Y$ on $\mathcal{R}(X) \times \mathcal{R}(Y)$, and let $h : \mathcal{R}(X) \times \mathcal{R}(Y) \rightarrow \mathbb{R}$ be measurable. If either*

- (a) $h(X, Y) \geq 0$ almost surely, or
- (b) $\mathbb{E}[|h(X, Y)|] < \infty$,

then the expectation of $h(X, Y)$ can be written as an iterated integral against $f_{X,Y}$, with the order of integration exchangeable:

$$\begin{aligned} \mathbb{E}[h(X, Y)] &= \int_{\mathcal{R}(X)} \left(\int_{\mathcal{R}(Y)} h(x, y) f_{X,Y}(x, y) d\mu_Y(y) \right) d\mu_X(x) \\ &= \int_{\mathcal{R}(Y)} \left(\int_{\mathcal{R}(X)} h(x, y) f_{X,Y}(x, y) d\mu_X(x) \right) d\mu_Y(y). \end{aligned}$$

The choice of reference measures covers three cases:

- **Both continuous:** $\mu_X = \mu_Y = \text{Lebesgue measure}$; $f_{X,Y}$ is the joint probability density function (PDF), and $\int g(x) d\mu_X(x) = \int g(x) dx$.
- **Both discrete:** $\mu_X = \mu_Y = \text{counting measure}$; $f_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$ is the joint probability mass function (PMF), and $\int g(x) d\mu_X(x) = \sum_{x \in \mathcal{R}(X)} g(x)$.
- **Mixed** (one continuous, one discrete): one reference measure is Lebesgue and the other is counting; $f_{X,Y}(x, y) = f_{X|Y}(x \mid y) \mathbb{P}(Y = y)$ (or $\mathbb{P}(X = x \mid Y = y) f_Y(y)$ if X is discrete and Y continuous), and the iterated integrals combine an ordinary integral with a sum.

i Proof

Proof. Apply Fubini–Tonelli^a with $\mu_1 = \mu_X$ and $\mu_2 = \mu_Y$ to the integrand $h(x, y) f_{X,Y}(x, y)$ on $\mathcal{R}(X) \times \mathcal{R}(Y)$. Lebesgue measure and counting measure on a countable set are each σ -finite, so $\mu_X \otimes \mu_Y$ is σ -finite in all three cases. The relevant hypothesis is (a) when $h \geq 0$ and (b) when $\mathbb{E}[|h(X, Y)|] < \infty$. Independence is not required. When X and Y are independent, $f_{X,Y}(x, y) = f_X(x) f_Y(y)$ (or $\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x) \mathbb{P}(Y = y)$ in the discrete case), and the iterated integrals factor into separate integrals over the marginals. $\square$

^a[math-prereqs.qmd#thm-fubini-tonelli](#)

⁴[math-prereqs.qmd#thm-fubini](#)

⁵[math-prereqs.qmd#thm-fubini-tonelli](#)

⁶[math-prereqs.qmd#thm-fubini-tonelli](#)

Exm

Example 3.3 (Expectation of a product of independent variables). Let $X \sim \text{Uniform}(0, 1)$ and $Y \sim \text{Uniform}(0, 2)$, independently distributed. Compute $E[XY]$.

We apply Corollary 3.1 (both-continuous case) with $h(x, y) = xy$. Since X and Y are independent with densities $f_X(x) = 1$ on $[0, 1]$ and $f_Y(y) = \frac{1}{2}$ on $[0, 2]$, the joint density factors as $f_{X,Y}(x, y) = f_X(x) f_Y(y) = \frac{1}{2}$, and $\mu_X = \mu_Y = \text{Lebesgue measure}$:

$$\begin{aligned} E[XY] &= \int_0^1 \left(\int_0^2 xy \cdot \frac{1}{2} dy \right) dx \\ &= \int_0^1 x \left(\frac{1}{2} \int_0^2 y dy \right) dx \\ &= \int_0^1 x \cdot \frac{1}{2} \cdot \left[\frac{y^2}{2} \right]_0^2 dx \\ &= \int_0^1 x \cdot \frac{1}{2} \cdot 2 dx \\ &= \int_0^1 x dx \\ &= \frac{1}{2} \end{aligned}$$

As a check: $E[X] = \frac{1}{2}$, $E[Y] = 1$, and $E[X] E[Y] = \frac{1}{2}$.

Exm

Example 3.4 (When independence fails: a counterexample). Correctly applying Corollary 3.1 requires the *actual* joint density $f_{X,Y}$ — not the product of marginals $f_X(x) f_Y(y)$, which is valid only when X and Y are independent. Using the wrong joint density gives the wrong answer.

Let $X \sim \text{Uniform}(0, 1)$ and set $Y = X$ (so X and Y are perfectly correlated and **not** independent).

True expectation:

$$E[XY] = E[X \cdot X] = E[X^2] = \int_0^1 x^2 dx = \frac{1}{3}$$

Erroneously applying the product-measure formula:

Note that Fubini–Tonelli’s own conditions still hold here ($h(x, y) = xy$ is nonnegative and integrable), so the error is not a failure of Fubini–Tonelli^a. Rather, the error is using the *wrong measure*: the joint distribution of (X, X) is concentrated on the diagonal $\{(x, x) : x \in [0, 1]\} \subset [0, 1]^2$, which has Lebesgue measure zero in $\mathbb{R}^2$. The joint distribution is therefore **not** absolutely continuous with respect to two-dimensional Lebesgue measure, so **no joint density** $f_{X,Y}$ **on** $[0, 1]^2$ **exists**, which is the reference density Corollary 3.1 requires.

The calculation below is what someone would *erroneously* write if they assumed independence and used $f_X(x) f_Y(y)$ as a “joint density” — a function that does not in fact correspond to the joint distribution of (X, X) . The marginals $X \sim \text{Uniform}(0, 1)$ and $Y \sim \text{Uniform}(0, 1)$ do have densities $f_X = f_Y = 1$, but the *product* $f_X(x) f_Y(y) = 1$ on $[0, 1]^2$ is the density of an *independent* pair, not of (X, X) :

$$\begin{aligned}
\int_0^1 \int_0^1 xy \cdot f_X(x) \cdot f_Y(y) \, dy \, dx &= \int_0^1 \int_0^1 xy \, dy \, dx \\
&= \int_0^1 x \left(\int_0^1 y \, dy \right) \, dx \\
&= \int_0^1 x \cdot \frac{1}{2} \, dx \\
&= \frac{1}{4}
\end{aligned}$$

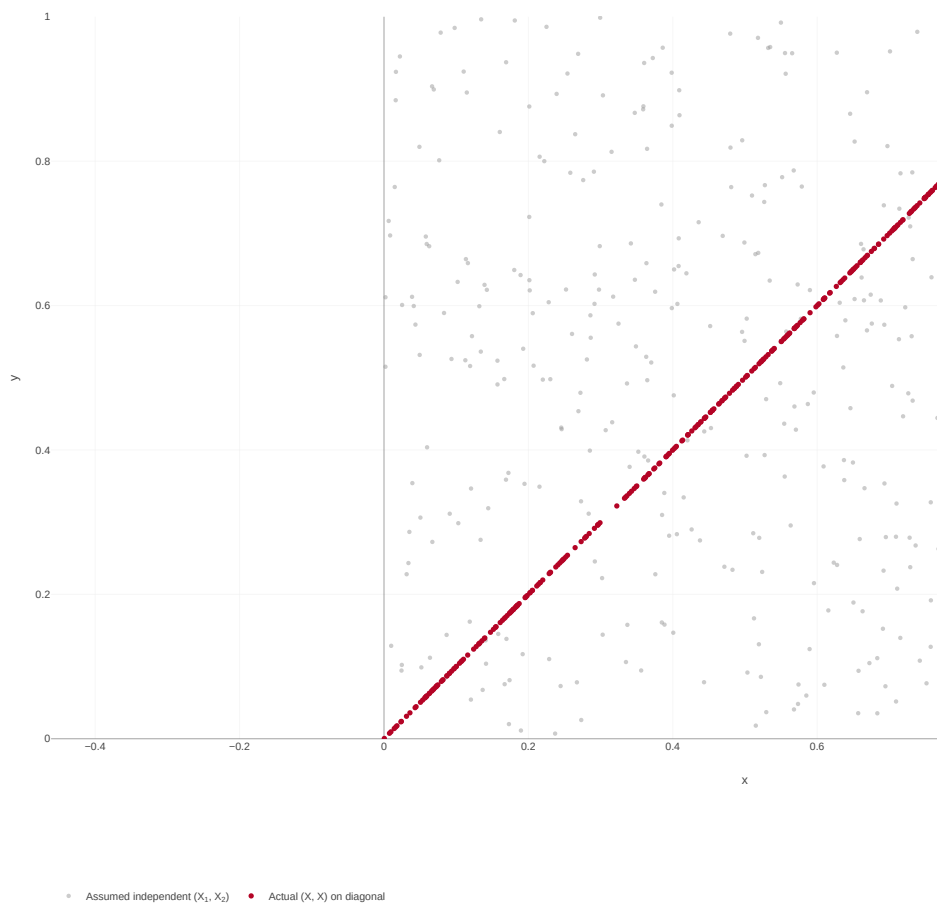
This recovers $E[XY]$ for *independent* uniforms ($\frac{1}{4}$), not $E[XX]$ for the perfectly correlated pair ($\frac{1}{3}$). The lesson is that Corollary 3.1 requires the *actual* joint density $f_{X,Y}$. For independent (X,Y) , this factors as $f_X(x)f_Y(y)$; for dependent (X,Y) , $f_{X,Y}$ need not factor — and for (X,X) , no joint density on $\mathbb{R}^2$ exists at all, so Corollary 3.1 simply does not apply.

```

set.seed(204)
n <- 400
x_dep <- runif(n)
y_dep <- x_dep
x_ind <- runif(n)
y_ind <- runif(n)

plotly::plot_ly() |>
  plotly::add_trace(
    type = "scatter", mode = "markers",
    x = x_ind, y = y_ind,
    name = "Assumed independent (X<sub>1</sub>, X<sub>2</sub>)",
    marker = list(size = 5, color = "#999999", opacity = 0.5)
  ) |>
  plotly::add_trace(
    type = "scatter", mode = "markers",
    x = x_dep, y = y_dep,
    name = "Actual (X, X) on diagonal",
    marker = list(size = 6, color = "#b40426")
  ) |>
  plotly::layout(
    xaxis = list(title = "x", range = c(0, 1), scaleanchor = "y"),
    yaxis = list(title = "y", range = c(0, 1)),
    legend = list(orientation = "h", y = -0.2)
  )

```



Exm

Example 3.5 (Both-continuous case: joint PDF on a non-rectangular support). Let (X, Y) have joint density $f_{X,Y}(x, y) = 2$ for $0 \leq x \leq y \leq 1$ (and 0 otherwise). Compute $E[X + Y]$. By Corollary 3.1:

$$\begin{aligned} E[X + Y] &= \int_0^1 \int_0^y (x + y) \cdot 2 \, dx \, dy \\ &= 2 \int_0^1 \left[\frac{x^2}{2} + xy \right]_{x=0}^{x=y} dy \\ &= 2 \int_0^1 \left(\frac{y^2}{2} + y^2 \right) dy \\ &= 2 \int_0^1 \frac{3y^2}{2} dy \\ &= 3 \int_0^1 y^2 dy \\ &= 3 \cdot \frac{1}{3} \\ &= 1 \end{aligned}$$

```

n_grid <- 51
x_seq <- seq(0, 1, length.out = n_grid)
y_seq <- seq(0, 1, length.out = n_grid)

z_mat <- outer(x_seq, y_seq, function(x, y) {
  z <- rep(2, length(x))
  z[x > y] <- NA
  z
})

plotly::plot_ly(x = ~x_seq, y = ~y_seq, z = ~t(z_mat)) |>
  plotly::add_surface(showscale = FALSE) |>
  plotly::layout(scene = list(
    xaxis = list(title = "x"),
    yaxis = list(title = "y"),
    zaxis = list(title = "f(x, y)", range = c(0, 2.5)),
    camera = list(eye = list(x = 1.6, y = -1.6, z = 0.8))
  ))

```

Exm

Example 3.6 (Both-discrete case: joint PMF). Let (X, Y) be discrete with joint probability mass function:

	$Y = 0$	$Y = 1$
$X = 0$	0.2	0.3
$X = 1$	0.1	0.4

Compute $E[X + Y]$ using Corollary 3.1 with $\mu_X = \mu_Y = \text{counting measure}$ and $h(x, y) = x + y$. By Corollary 3.1 (both-discrete case):

$$\begin{aligned} E[X + Y] &= \sum_{x \in \{0,1\}} \sum_{y \in \{0,1\}} (x + y) P(X = x, Y = y) \\ &= (0+0)(0.2) + (0+1)(0.3) + (1+0)(0.1) + (1+1)(0.4) \\ &= 0 + 0.3 + 0.1 + 0.8 \\ &= 1.2 \end{aligned}$$

As a check: $E[X] = 0(0.5) + 1(0.5) = 0.5$ and $E[Y] = 0(0.3) + 1(0.7) = 0.7$, so $E[X + Y] = E[X] + E[Y] = 1.2$.

Note that X and Y are **not** independent here: $P(X = 0, Y = 0) = 0.2 \neq 0.15 = P(X = 0) P(Y = 0)$. Corollary 3.1 applies regardless, since it requires only the *actual* joint mass function, not independence.

```

x_labs <- c("X=0", "X=0", "X=1", "X=1")
y_labs <- c("Y=0", "Y=1", "Y=0", "Y=1")
probs <- c(0.2, 0.3, 0.1, 0.4)

plotly::plot_ly(
  x = ~y_labs, y = ~probs, color = ~x_labs,
  colors = c("steelblue", "tomato"),
  type = "bar"
) |>
plotly::layout(
  barmode = "group",
  xaxis = list(title = "Y"),
  yaxis = list(title = "P(X = x, Y = y)", range = c(0, 0.5)),
  legend = list(title = list(text = "X value"))
)

```

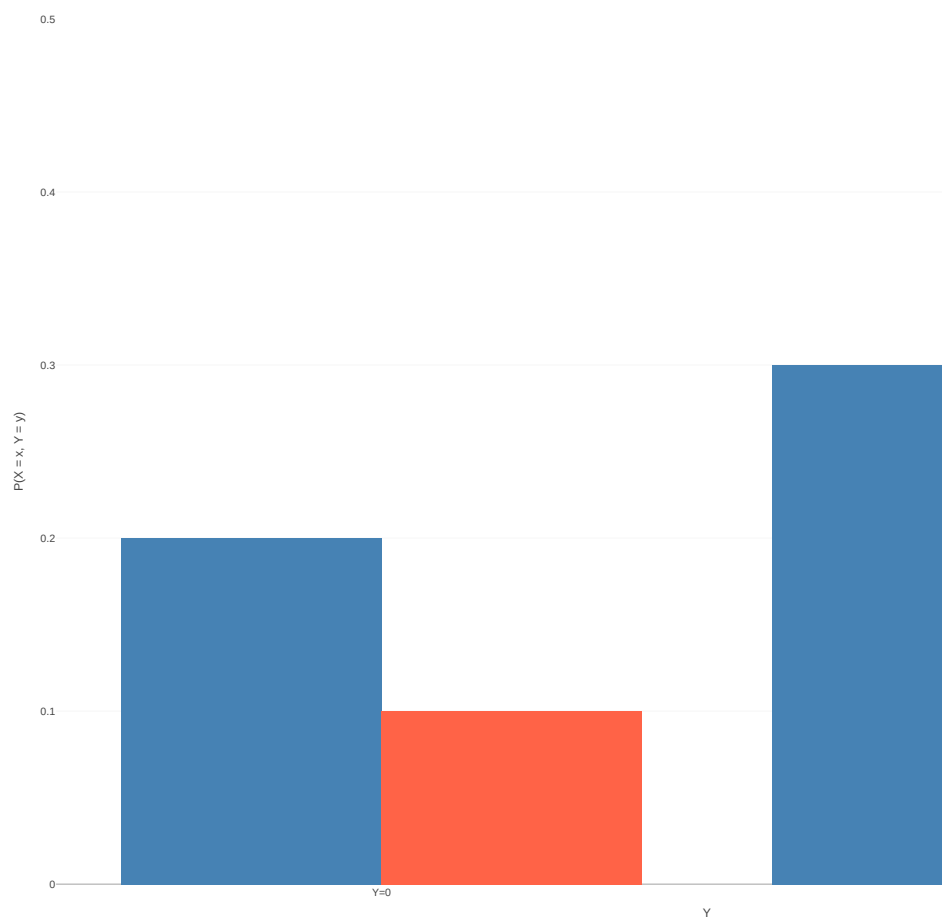


Figure 6: Joint probability mass function $P(X = x, Y = y)$. Marginal totals: $P(X = 0) = 0.5$, $P(X = 1) = 0.5$, $P(Y = 0) = 0.3$, $P(Y = 1) = 0.7$.

Example 3.7 (Mixed case: one continuous variable, one discrete variable). Let $Y \sim \text{Bernoulli}(0.6)$ and, given $Y = y$, let $X \mid Y = y \sim \text{Uniform}(0, y + 1)$. Compute $E[X]$ using Corollary 3.1 with $\mu_X = \text{Lebesgue measure}$, $\mu_Y = \text{counting measure}$, and $h(x, y) = x$. The joint density w.r.t. Lebesgue $\times$ counting measure is $f_{X,Y}(x, y) = f_{X|Y}(x \mid y) P(Y = y)$:

$$\begin{aligned} f_{X,Y}(x, 0) &= 1 \cdot 0.4 = 0.4 && \text{for } x \in [0, 1]; \\ f_{X,Y}(x, 1) &= \tfrac{1}{2} \cdot 0.6 = 0.3 && \text{for } x \in [0, 2]. \end{aligned}$$

By Corollary 3.1 (mixed case):

$$\begin{aligned} E[X] &= \sum_{y \in \{0,1\}} \int_0^{y+1} x f_{X,Y}(x, y) dx \\ &= \int_0^1 x \cdot 0.4 dx + \int_0^2 x \cdot 0.3 dx \\ &= 0.4 \cdot \frac{1}{2} + 0.3 \cdot 2 \\ &= 0.2 + 0.6 = 0.8 \end{aligned}$$

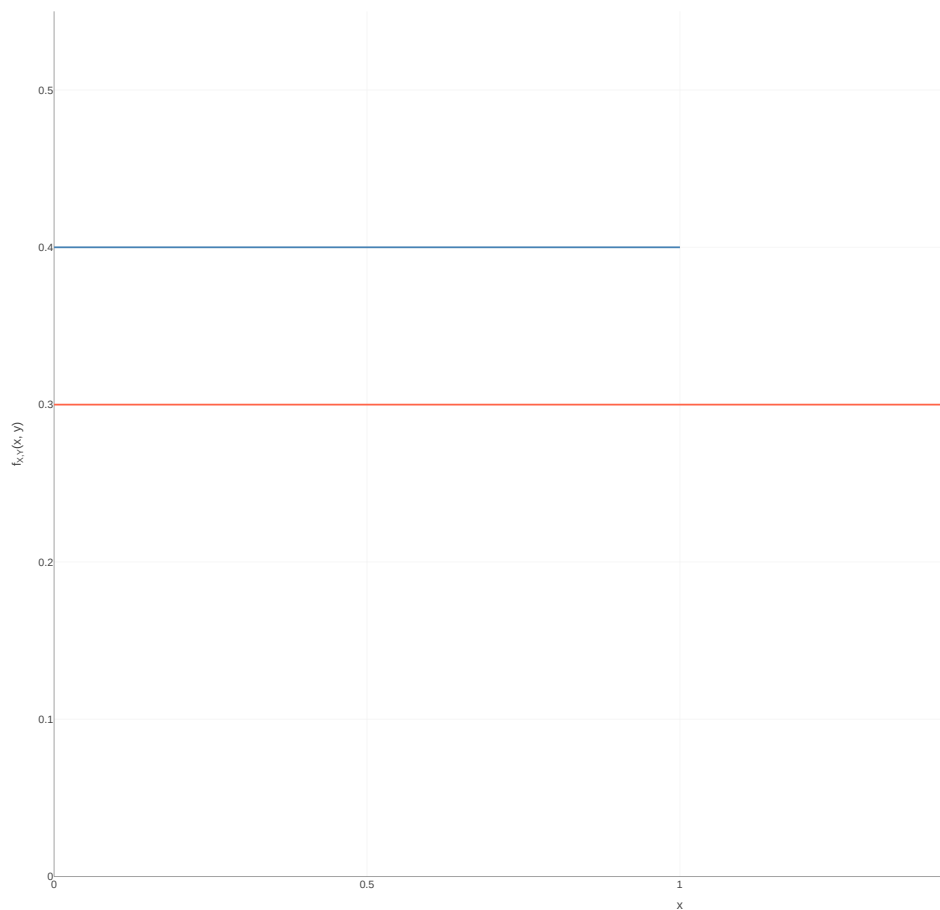
As a check using the law of total expectation: $E[X \mid Y = 0] = \frac{1}{2}$ and $E[X \mid Y = 1] = 1$, so $E[X] = \frac{1}{2}(0.4) + 1(0.6) = 0.2 + 0.6 = 0.8$.

```

x_fine <- seq(0, 2, by = 0.005)
df <- data.frame(
  x = c(x_fine[x_fine <= 1], x_fine),
  density = c(rep(0.4, sum(x_fine <= 1)), rep(0.3, length(x_fine))),
  label = c(
    rep("Y = 0 (P = 0.4)", sum(x_fine <= 1)),
    rep("Y = 1 (P = 0.6)", length(x_fine))
  )
)

plotly::plot_ly(
  df, x = ~x, y = ~density, color = ~label,
  colors = c("steelblue", "tomato")
) |>
plotly::add_lines() |>
plotly::layout(
  xaxis = list(title = "x"),
  yaxis = list(title = "fX,Y(x, y)", range = c(0, 0.55)),
  legend = list(title = list(text = "Y value"))
)

```



Theorem 3.6 (Law of iterated expectations). *For any two random variables X and Y :*

$$E[Y] = E[E[Y | X]]$$

*Alternate names for this identity include: the **tower rule**, the **tower property**, the **law of total expectation**, and the **smoothing theorem**.*

i Proof

Proof. Discrete case. When X and Y are discrete, applying Definition 3.5 to $E[E[Y | X]]$ and then the law of total probability (Theorem 1.4) applied to the countable partition $\{X = x : x \in \mathcal{R}(X)\}$:

$$\begin{aligned} E[E[Y | X]] &= \sum_{x \in \mathcal{R}(X)} E[Y | X = x] \cdot P(X = x) \\ &= \sum_{x \in \mathcal{R}(X)} \left(\sum_{y \in \mathcal{R}(Y)} y \cdot P(Y = y | X = x) \right) \cdot P(X = x) \\ &= \sum_{y \in \mathcal{R}(Y)} y \cdot \sum_{x \in \mathcal{R}(X)} P(Y = y | X = x) \cdot P(X = x) \\ &= \sum_{y \in \mathcal{R}(Y)} y \cdot P(Y = y) \\ &= E[Y] \end{aligned}$$

Continuous case. When X and Y are continuous, applying Definition 3.5 to $E[E[Y | X]]$ and then using Definition 3.6 for $E[Y | X = x]$:

$$\begin{aligned} E[E[Y | X]] &= \int_{x \in \mathcal{R}(X)} E[Y | X = x] \cdot p(X = x) dx \\ &= \int_{x \in \mathcal{R}(X)} \left(\int_{y \in \mathcal{R}(Y)} y \cdot p(Y = y | X = x) dy \right) \cdot p(X = x) dx \\ &= \int_{y \in \mathcal{R}(Y)} y \cdot \left(\int_{x \in \mathcal{R}(X)} p(Y = y | X = x) \cdot p(X = x) dx \right) dy \\ &= \int_{y \in \mathcal{R}(Y)} y \cdot p(Y = y) dy \\ &= E[Y] \end{aligned}$$

where the third equality exchanges the order of integration by hypothesis (b) of Fubini–Tonelli^a (the absolute-integrability case, **Fubini’s theorem**); this requires $E[|Y|] < \infty$, which is implicit in $E[Y]$ being defined, and the fourth equality uses $\int_x p(Y = y | X = x) \cdot p(X = x) dx = \int_x p(X = x, Y = y) dx = p(Y = y)$ (marginalization of the joint density). $\square$

^a[math-prereqs.qmd#thm-fubini-tonelli](#)

Theorem 3.7 (Conditional law of iterated expectations). *For random variables X , Y , and Z :*

$$E[Y | Z] = E[E[Y | X, Z] | Z]$$

This is the tower rule applied conditionally on Z .

i Proof

Proof. For each fixed value z with positive probability or density:

Discrete case. Conditioning on $Z = z$, and applying the law of total probability to the partition $\{X = x : x \in \mathcal{R}(X)\}$ under the conditional distribution given $Z = z$:

$$\begin{aligned} E[E[Y | X, Z] | Z = z] &= \sum_{x \in \mathcal{R}(X)} E[Y | X = x, Z = z] \cdot P(X = x | Z = z) \\ &= E[Y | Z = z] \end{aligned}$$

Continuous case. Conditioning on $Z = z$, and integrating over X under the conditional density $p(X = x | Z = z)$:

$$\begin{aligned} E[E[Y | X, Z] | Z = z] &= \int_{x \in \mathcal{R}(X)} E[Y | X = x, Z = z] \cdot p(X = x | Z = z) dx \\ &= E[Y | Z = z] \end{aligned}$$

Therefore, as random variables of Z , $E[Y | Z] = E[E[Y | X, Z] | Z]$. □

Exm

Example 3.8 (Marginal expectation from conditional expectations). Suppose X is a binary random variable indicating treatment assignment ($X = 1$ treated, $X = 0$ control), with $P(X = 1) = 0.5$, and suppose the outcome Y has conditional expectations:

$$E[Y | X = 1] = 10, \quad E[Y | X = 0] = 6$$

By the law of iterated expectations (Theorem 3.6):

$$\begin{aligned} E[Y] &= E[E[Y | X]] \\ &= E[Y | X = 1] \cdot P(X = 1) + E[Y | X = 0] \cdot P(X = 0) \\ &= 10 \cdot 0.5 + 6 \cdot 0.5 \\ &= 5 + 3 \\ &= 8 \end{aligned}$$

Definition 3.7 (Expectation of a random matrix). For a random matrix $\mathbf{A}$ of size $m \times n$ with (i, j) -th element A_{ij} , the **expectation** $E \mathbf{A}$ is the $m \times n$ matrix whose (i, j) -th element is $E[A_{ij}]$:

$$E \mathbf{A} \stackrel{\text{def}}{=} \begin{pmatrix} E[A_{11}] & E[A_{12}] & \cdots & E[A_{1n}] \\ E[A_{21}] & E[A_{22}] & \cdots & E[A_{2n}] \\ \vdots & \vdots & \ddots & \vdots \\ E[A_{m1}] & E[A_{m2}] & \cdots & E[A_{mn}] \end{pmatrix}$$

In other words, expectation is applied **element-wise** to a random matrix.

3.5 Deviation, error, and noise

Definition 3.8 (Deviation). A **deviation** is the difference between a value and a reference value. For any quantity z and reference value r :

$$z - r$$

In probability and statistics, “deviation” often means deviation from a [population mean](#). For a random variable Y :

$$Y - E[Y]$$

See: Wikipedia: Deviation (statistics)^a

^a[https://en.wikipedia.org/wiki/Deviation_\(statistics\)](https://en.wikipedia.org/wiki/Deviation_(statistics))

Definition 3.9 (Deviation from a population or subpopulation mean). In probabilistic models, we call this quantity a **deviation from a mean**. It is often also called an **error** or **noise term** in other sources. For the random variable Y , define the [deviation](#) from its mean as:

$$e(Y) \stackrel{\text{def}}{=} Y - E[Y]$$

For a realized observation y :

$$e(y) \stackrel{\text{def}}{=} y - E[Y]$$

In regression settings, the reference mean is often conditional on covariates: $e(y_i) \stackrel{\text{def}}{=} y_i - E[Y_i | X_i]$.

In this course, we prefer “deviation” for this mean-deviation quantity. The terms “error” and “noise” are common aliases. We use “residual” (defined in the Linear regression chapter^a) for deviations from fitted values. For notation in this course, we use $e(\cdot)$ for these model/data deviations, and reserve $\varepsilon(\cdot)$ for estimator-to-estimand deviations (see Estimation^b).

See:

- Wikipedia: Errors and residuals^c
- Wikipedia: Deviation (statistics)^d
- Wikipedia: Linear regression — Notation and terminology^e

^a[Linear-models-overview.qmd#def-resid-fitted](#)

^b[estimation.qmd#def-estimation-error](#)

^chttps://en.wikipedia.org/wiki/Errors_and_residuals

^d[https://en.wikipedia.org/wiki/Deviation_\(statistics\)](https://en.wikipedia.org/wiki/Deviation_(statistics))

^ehttps://en.wikipedia.org/wiki/Linear_regression#Notation_and_terminology

3.6 Variance and related characteristics

Definition 3.10 (Variance). The variance of a random variable X is the [expectation](#) of the squared [deviation from the mean](#); that is:

$$\text{Var}(X) \stackrel{\text{def}}{=} E[e(X)^2]$$

Theorem 3.8 (Variance as expected squared deviation from the mean).

$$\text{Var}(X) = E[(X - E[X])^2]$$

Proof

Proof. Substituting the definition of $e(X)$ from Definition 3.9 into Definition 3.10:

$$\text{Var}(X) \stackrel{\text{def}}{=} \mathbb{E}[e(X)^2] = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

□

Theorem 3.9 (Simplified expression for variance).

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Proof

Proof. By linearity of expectation, we have:

$$\begin{aligned} \text{Var}(X) &\stackrel{\text{def}}{=} \mathbb{E}[e(X)^2] \\ &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2X\mathbb{E}[X]] + \mathbb{E}[(\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}[X])^2 \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

□

Theorem 3.10 (Law of total variance). *For random variables X and Y :*

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y | X)] + \text{Var}(\mathbb{E}[Y | X])$$

where $\text{Var}(Y | X) \stackrel{\text{def}}{=} \mathbb{E}[(Y - \mathbb{E}[Y | X])^2 | X]$.

*Alternate names include: the **conditional variance formula**, **Eve's law**, and the **variance decomposition formula**.*

i Proof

Proof. Write $Y - \mathbb{E}[Y] = (Y - \mathbb{E}[Y | X]) + (\mathbb{E}[Y | X] - \mathbb{E}[Y])$. Then:

$$(Y - \mathbb{E}[Y])^2 = (Y - \mathbb{E}[Y | X])^2 + (\mathbb{E}[Y | X] - \mathbb{E}[Y])^2 + 2(Y - \mathbb{E}[Y | X])(\mathbb{E}[Y | X] - \mathbb{E}[Y])$$

Taking expectation:

$$\begin{aligned} \text{Var}(Y) &= \mathbb{E}[(Y - \mathbb{E}[Y | X])^2] + \mathbb{E}[(\mathbb{E}[Y | X] - \mathbb{E}[Y])^2] \\ &\quad + 2\mathbb{E}[(Y - \mathbb{E}[Y | X])(\mathbb{E}[Y | X] - \mathbb{E}[Y])] \end{aligned}$$

For the cross-term:

Discrete case.

$$\begin{aligned}
E[(Y - E[Y | X])(E[Y | X] - E[Y])] &= \sum_{x \in \mathcal{R}(X)} E[(Y - E[Y | X])(E[Y | X] - E[Y]) | X = x] \cdot P(X = x) \\
&= \sum_{x \in \mathcal{R}(X)} (E[Y | X = x] - E[Y]) \cdot E[Y - E[Y | X = x] | X = x] \cdot P(X = x) \\
&= 0
\end{aligned}$$

Continuous case.

$$\begin{aligned}
E[(Y - E[Y | X])(E[Y | X] - E[Y])] &= \int_{x \in \mathcal{R}(X)} E[(Y - E[Y | X])(E[Y | X] - E[Y]) | X = x] \cdot p(X = x) dx \\
&= \int_{x \in \mathcal{R}(X)} (E[Y | X = x] - E[Y]) \cdot E[Y - E[Y | X = x] | X = x] \cdot p(X = x) dx \\
&= 0
\end{aligned}$$

Therefore:

$$\begin{aligned}
\text{Var}(Y) &= E[(Y - E[Y | X])^2] + E[(E[Y | X] - E[Y])^2] \\
&= E[\text{Var}(Y | X)] + \text{Var}(E[Y | X])
\end{aligned}$$

□

Definition 3.11 (Precision). The **precision** of a random variable X , often denoted $\tau(X)$, τ_X , or shorthand as τ , is the inverse of that random variable's **variance**; that is:

$$\tau(X) \stackrel{\text{def}}{=} (\text{Var}(X))^{-1}$$

Definition 3.12 (Standard deviation). The standard deviation of a random variable X is the square-root of the **variance** of X :

$$\text{SD}(X) \stackrel{\text{def}}{=} \sqrt{\text{Var}(X)}$$

Definition 3.13 (Covariance). For any two one-dimensional random variables, X, Y :

$$\text{Cov}(X, Y) \stackrel{\text{def}}{=} E[(X - E[X])(Y - E[Y])]$$

Theorem 3.11 (Alternative formula for covariance).

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

Theorem 3.12 (Law of total covariance). *For random variables X, Y , and Z :*

$$\text{Cov}(Y, Z) = E[\text{Cov}(Y, Z | X)] + \text{Cov}(E[Y | X], E[Z | X])$$

where $\text{Cov}(Y, Z | X) \stackrel{\text{def}}{=} E[(Y - E[Y | X])(Z - E[Z | X]) | X]$.

Alternate names include: the **covariance decomposition formula** and the **conditional covariance formula**.

i Proof

Proof. Write:

$$\begin{aligned} Y - E[Y] &= (Y - E[Y | X]) + (E[Y | X] - E[Y]) \\ Z - E[Z] &= (Z - E[Z | X]) + (E[Z | X] - E[Z]) \end{aligned}$$

Then:

$$\begin{aligned} \text{Cov}(Y, Z) &= E[(Y - E[Y])(Z - E[Z])] \\ &= E[(Y - E[Y | X])(Z - E[Z | X])] \\ &\quad + E[(Y - E[Y | X])(E[Z | X] - E[Z])] \\ &\quad + E[(E[Y | X] - E[Y])(Z - E[Z | X])] \\ &\quad + E[(E[Y | X] - E[Y])(E[Z | X] - E[Z])] \end{aligned}$$

For the two mixed terms:

Discrete case.

$$\begin{aligned} E[(Y - E[Y | X])(E[Z | X] - E[Z])] &= \sum_{x \in \mathcal{R}(X)} E[(Y - E[Y | X])(E[Z | X] - E[Z]) | X = x] \cdot P(X = x) \\ &= \sum_{x \in \mathcal{R}(X)} (E[Z | X = x] - E[Z]) \cdot E[Y - E[Y | X = x] | X = x] \cdot P(X = x) \\ &= 0 \end{aligned}$$

and similarly:

$$E[(E[Y | X] - E[Y])(Z - E[Z | X])] = 0.$$

Continuous case.

$$\begin{aligned} E[(Y - E[Y | X])(E[Z | X] - E[Z])] &= \int_{x \in \mathcal{R}(X)} E[(Y - E[Y | X])(E[Z | X] - E[Z]) | X = x] \cdot p(X = x) dx \\ &= \int_{x \in \mathcal{R}(X)} (E[Z | X = x] - E[Z]) \cdot E[Y - E[Y | X = x] | X = x] \cdot p(X = x) dx \\ &= 0 \end{aligned}$$

and similarly:

$$E[(E[Y | X] - E[Y])(Z - E[Z | X])] = 0.$$

Hence:

$$\begin{aligned} \text{Cov}(Y, Z) &= E[(Y - E[Y | X])(Z - E[Z | X])] + E[(E[Y | X] - E[Y])(E[Z | X] - E[Z])] \\ &= E[\text{Cov}(Y, Z | X)] + \text{Cov}(E[Y | X], E[Z | X]) \end{aligned}$$

□

Lemma 3.1 (The covariance of a variable with itself is its variance). *For any random variable X :*

$$\text{Cov}(X, X) = \text{Var}(X)$$

i Proof

Proof.

$$\begin{aligned}\text{Cov}(X, X) &= \text{E}[XX] - \text{E}[X] \text{E}[X] \\ &= \text{E}[X^2] - (\text{E}[X])^2 \\ &= \text{Var}(X)\end{aligned}$$

□

Definition 3.14 (Variance/covariance of a $p \times 1$ random vector). For a $p \times 1$ dimensional random vector $\tilde{X}$,

$$\begin{aligned}\text{Var}(\tilde{X}) &\stackrel{\text{def}}{=} \text{Cov}(\tilde{X}) \\ &\stackrel{\text{def}}{=} \text{E}\left[(\tilde{X} - \text{E}\tilde{X})(\tilde{X} - \text{E}\tilde{X})^\top\right]\end{aligned}$$

Theorem 3.13 (Elements of the variance-covariance matrix are pairwise covariances). *For a $p \times 1$ random vector $\tilde{X} = (X_1, \dots, X_p)^\top$, the (i, j) -th element of $\text{Var}(\tilde{X})$ is $\text{Cov}(X_i, X_j)$:*

$$\text{Var}(\tilde{X}) = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_p) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots & \text{Cov}(X_2, X_p) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \text{Cov}(X_p, X_2) & \cdots & \text{Var}(X_p) \end{pmatrix}$$

i Proof

Proof. Let $\mu_i = \text{E}[X_i]$ for $i = 1, \dots, p$, so $\text{E}\tilde{X} = (\mu_1, \dots, \mu_p)^\top$. By Definition 3.14:

$$\begin{aligned}
\text{Var}(\tilde{X}) &= \mathbb{E}[(\tilde{X} - \mathbb{E} \tilde{X})(\tilde{X} - \mathbb{E} \tilde{X})^\top] \\
&= \mathbb{E} \left[\begin{pmatrix} X_1 - \mu_1 \\ \vdots \\ X_p - \mu_p \end{pmatrix} (X_1 - \mu_1 \quad \cdots \quad X_p - \mu_p) \right] \\
&= \mathbb{E} \left[\begin{pmatrix} (X_1 - \mu_1)(X_1 - \mu_1) & \cdots & (X_1 - \mu_1)(X_p - \mu_p) \\ \vdots & \ddots & \vdots \\ (X_p - \mu_p)(X_1 - \mu_1) & \cdots & (X_p - \mu_p)(X_p - \mu_p) \end{pmatrix} \right] \\
&= \begin{pmatrix} \mathbb{E}[(X_1 - \mu_1)(X_1 - \mu_1)] & \cdots & \mathbb{E}[(X_1 - \mu_1)(X_p - \mu_p)] \\ \vdots & \ddots & \vdots \\ \mathbb{E}[(X_p - \mu_p)(X_1 - \mu_1)] & \cdots & \mathbb{E}[(X_p - \mu_p)(X_p - \mu_p)] \end{pmatrix} \\
&= \begin{pmatrix} \text{Cov}(X_1, X_1) & \cdots & \text{Cov}(X_1, X_p) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \cdots & \text{Cov}(X_p, X_p) \end{pmatrix} \\
&= \begin{pmatrix} \text{Var}(X_1) & \cdots & \text{Cov}(X_1, X_p) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \cdots & \text{Var}(X_p) \end{pmatrix}
\end{aligned}$$

where:

- the step from the third to fourth line uses Definition 3.7,
- the step from the fourth to fifth line uses Definition 3.13, and
- the last step uses Lemma 3.1.

□

Theorem 3.14 (Alternate expression for variance of a random vector).

$$\text{Var}(\tilde{X}) = \mathbb{E}[\tilde{X}\tilde{X}^\top] - (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top$$

i Proof

Proof.

$$\begin{aligned}
\text{Var}(\tilde{X}) &= \mathbb{E}[(\tilde{X} - \mathbb{E} \tilde{X})(\tilde{X} - \mathbb{E} \tilde{X})^\top] \\
&= \mathbb{E}[\tilde{X}\tilde{X}^\top - \tilde{X}(\mathbb{E} \tilde{X})^\top - (\mathbb{E} \tilde{X})\tilde{X}^\top + (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top] \\
&= \mathbb{E}[\tilde{X}\tilde{X}^\top] - (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top - (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top + (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top \\
&= \mathbb{E}[\tilde{X}\tilde{X}^\top] - (\mathbb{E} \tilde{X})(\mathbb{E} \tilde{X})^\top
\end{aligned}$$

□

Theorem 3.15 (Variance of a linear combination). *For any vector of random variables $\tilde{X} = (X_1, \dots, X_n)$ and corresponding vector of constants $\tilde{a} = (a_1, \dots, a_n)$, the variance of their linear combination is:*

$$\begin{aligned}
\text{Var}(\tilde{a} \cdot \tilde{X}) &= \text{Var}\left(\sum_{i=1}^n a_i X_i\right) \\
&= \tilde{a}^\top \text{Var}(\tilde{X}) \tilde{a} \\
&= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j)
\end{aligned}$$

i Proof

Proof. Left to the reader...

□

Corollary 3.2 (Variance of a sum of two random variables). *For any two random variables X and Y and scalars a and b :*

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2(a \cdot b) \text{Cov}(X, Y)$$

i Proof

Proof. Apply Theorem 3.15 with $n = 2$, $X_1 = X$, and $X_2 = Y$.
Or, see <https://statproofbook.github.io/P/var-lincomb.html>

□

Definition 3.15 (homoskedastic, heteroskedastic). A random variable Y is **homoskedastic** (with respect to covariates X) if the **variance** of Y does not vary with X :

$$\text{Var}(Y|X = x) = \sigma^2, \forall x$$

Otherwise it is **heteroskedastic**.

Definition 3.16 (Statistical independence). A set of random variables $X_1, \dots, X_n$ are **statistically independent** if their joint **probability** is equal to the product of their marginal **probabilities**:

$$\Pr(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \Pr(X_i = x_i)$$

💡 Tip

The symbol for independence, $\perp\!\!\!\perp$, is essentially just $\prod$ upside-down. So the symbol can remind you of its definition (Definition 3.16).

Definition 3.17 (Conditional independence). A set of random variables $Y_1, \dots, Y_n$ are **conditionally statistically independent** given a set of covariates $X_1, \dots, X_n$ if the joint **probability**

of the Y_i s given the X_i s is equal to the product of their marginal [probabilities](#):

$$\Pr(Y_1 = y_1, \dots, Y_n = y_n | X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \Pr(Y_i = y_i | X_i = x_i)$$

Definition 3.18 (Identically distributed). A set of random variables $X_1, \dots, X_n$ are **identically distributed** if they have the same range $\mathcal{R}(X)$ and if their marginal distributions $P(X_1 = x_1), \dots, P(X_n = x_n)$ are all equal to some shared distribution $P(X = x)$:

$$\forall i \in \{1 : n\}, \forall x \in \mathcal{R}(X) : P(X_i = x) = P(X = x)$$

Definition 3.19 (Conditionally identically distributed). A set of random variables $Y_1, \dots, Y_n$ are **conditionally identically distributed** given a set of covariates $X_1, \dots, X_n$ if $Y_1, \dots, Y_n$ have the same range $\mathcal{R}(X)$ and if the distributions $P(Y_i = y_i | X_i = x_i)$ are all equal to the same distribution $P(Y = y | X = x)$:

$$P(Y_i = y | X_i = x) = P(Y = y | X = x)$$

Definition 3.20 (Independent and identically distributed). A set of random variables $X_1, \dots, X_n$ are **independent and identically distributed** (shorthand: “ X_i iid”) if they are [statistically independent](#) and [identically distributed](#).

Definition 3.21 (Conditionally independent and identically distributed). A set of random variables $Y_1, \dots, Y_n$ are **conditionally independent and identically distributed** (shorthand: “ $Y_i | X_i$ ciid” or just “ $Y_i | X_i$ iid”) given a set of covariates $X_1, \dots, X_n$ if $Y_1, \dots, Y_n$ are [conditionally independent](#) given $X_1, \dots, X_n$ and $Y_1, \dots, Y_n$ are [conditionally identically distributed](#) given $X_1, \dots, X_n$.

3.7 The Central Limit Theorem

The sum of many independent or nearly-independent random variables with small variances (relative to the number of RVs being summed) produces bell-shaped distributions.

For example, consider the sum of five dice (Figure 8).

```
library(dplyr)
dist =
  expand.grid(1:6, 1:6, 1:6, 1:6, 1:6) |>
  rowwise() |>
  mutate(total = sum(c_across(everything())) |>
  ungroup() |>
  count(total) |>
  mutate(`p(X=x)` = n/sum(n))

library(ggplot2)

dist |>
  ggplot() +
```

```

aes(x = total, y = `p(X=x)`) +
geom_col() +
xlab("sum of dice (x)") +
ylab("Probability of outcome, Pr(X=x)") +
expand_limits(y = 0)

```

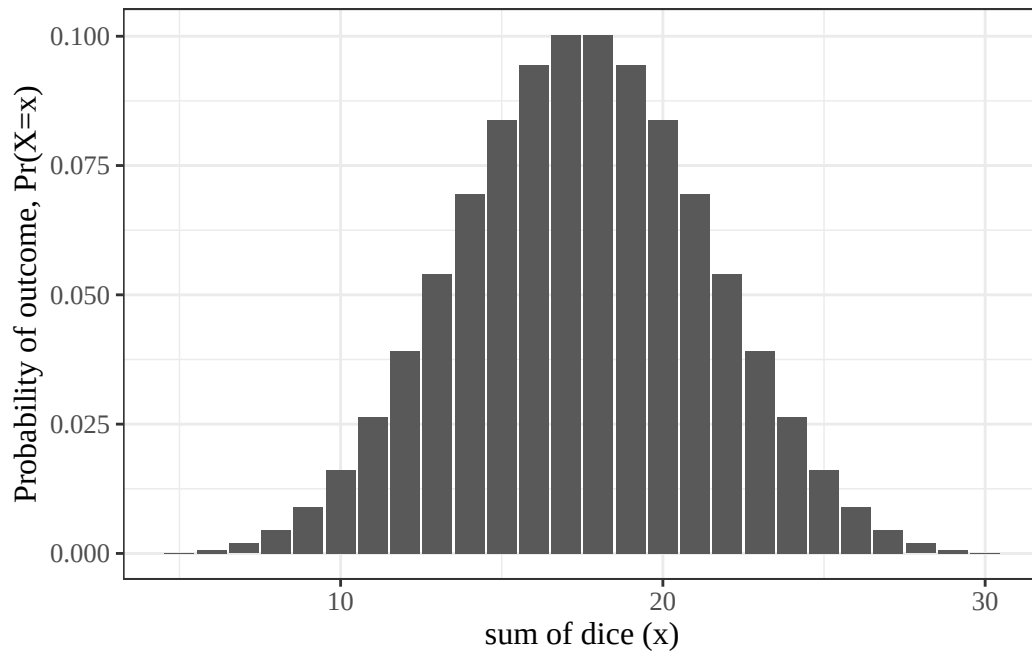


Figure 8: Distribution of the sum of five dice

In comparison, the outcome of just one die is not bell-shaped (Figure 9).

```

library(dplyr)
dist =
  expand.grid(1:6) |>
  rowwise() |>
  mutate(total = sum(c_across(everything())) |>
  ungroup() |>
  count(total) |>
  mutate(`p(X=x)` = n/sum(n))

library(ggplot2)

dist |>
  ggplot() +
  aes(x = total, y = `p(X=x)`) +
  geom_col() +
  xlab("sum of dice (x)") +
  ylab("Probability of outcome, Pr(X=x)") +
  expand_limits(y = 0)

```

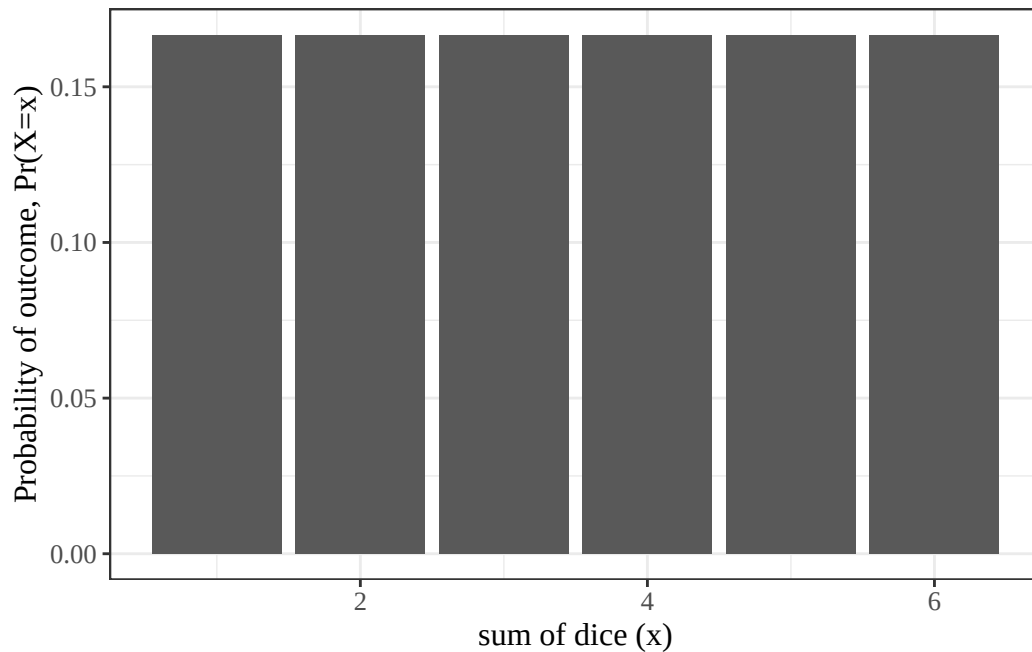


Figure 9: Distribution of the outcome of one die

What distribution does a single die have?

Answer: discrete uniform on 1:6.

4 Additional resources

- Miller (2017)

References

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